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# MILITARY SPECIFICATION

## SWEEPER, MAGNET, SELF-PROPELLED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers a sweeper consisting essentially of a tractor, equipped with an electromagnet assembly, hydraulically operated magnet height controls, and a 5 kilowatt (kW), direct current (dc), generator driven by the tractor engine.

## 2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

KKK-T-640 - Tractor, Wheeled, Agricultural, (Gasoline-Engine Driven, Diesel-Engine Driven, LP Gas-Engine Driven).

MILITARY

MIL-C-104 - Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116 - Preservation, Methods of.
MIL-T-704 - Treatment and Painting of Material.
MIL-T-3351 - Tractor, Full-Tracked, Low Speed; Tractor Wheeled, Agricultural; and Tractor Wheeled, Industrial: and Their Attachments, Packaging of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3825

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MIL-E-16298 - Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of.
MIL-C-46168 - Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-C-52950 - Crates, Wood, Open and Covered.

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-781 - Reliability Tests: Exponential Distribution.
MIL-STD-810 - Environmental Test Methods.
MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MS500048 - Towbar, Motor Vehicle: Light, Medium and Heavy Duty.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

DEPARTMENT OF LABOR (DoL)
OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Occupational Safety and Health Standards.

(Application for copies of DoL should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D709 - Specifications for Laminated Thermosetting Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)



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NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

MG1 - Motors and Generators.
MG2 - Safety Standard for Construction and Guide for Selection, Installation and Use of Electric Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATION, INC.

T&RA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

* 3.1 Standard commercial product. The sweeper shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the sweeper being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.2 First article (preproduction model). When specified (see 6.2), the contractor shall furnish a sweeper for first article inspection and approval (see 4.2.1 and 6.4), within the time frame specified. Examination and tests shall be as specified in section 4, and unless otherwise specified herein, all



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examinations and tests shall be conducted by the contractor subject to surveillance and approval by the Government. When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests, as specified (see 6.2).

* 3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.4 Identical items. All sweepers of the same classification furnished under a specific contract shall be physically and mechanically identical. This requirement includes parts, assemblies, components, and accessories. No deviations will be acceptable without written approval of the contracting officer.

3.5 Design and construction.

3.5.1 Reliability. The sweeper shall be so designed and constructed as to insure the highest degree of operational reliability under all service conditions. The "specified" mean-time-between-failure shall be 60 hours when tested in accordance with MIL-STD-781 as specified in 4.4.10.

3.5.2 Construction. Sweeper shall be so designed and constructed, that normal adjustments and repairs can be readily accomplished by means of general purpose tools with a minimum removal of disturbance of other elements of the unit. Covers or plates that must be removed for component adjustment or for parts removal, shall be equipped with substantial, quick-disconnect fastenings.

* 3.5.3 Safety and human factors. All exposed parts of the equipment which are subject to extremes of temperature, or which are energized electrically, and all moving parts which are of such nature or so located as to be a hazard to operating or maintenance personnel, shall be insulated, fully enclosed, or guarded. Protective devices shall not impair the operating functions. The sweeper shall conform to all Federal regulations regarding air pollution, rollover protection, and all other health and safety regulations as required by OSHA which are applicable for this type equipment. The characteristics of the sweeper shall provide for operation by personnel ranging from the small man (5th percentile) clothed, through the large man (95th percentile) arctic-clothed, in accordance with MIL-STD-1472.

3.5.3.1 Sound level. The sound level at the operator's station shall not exceed 85 decibels absolute (dB(A)) (re 20 megapascals (MPa)) when measured in accordance with SAE J919. The exterior sound level shall not exceed 88 dB(A) (re 20 MPa) when measured in accordance with SAE J88.



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3.5.4 Dissimilar metals. Intimate contact which can be expected to cause galvanic corrosion shall be avoided. When such contact can not be avoided, an interposing insulating material shall be provided to minimize the corrosive effect.

3.6 Performance.

3.6.1 Sweeping speed. The sweeper shall have a minimum sweeping speed of not more than 2.5 miles per hour (mph), and a maximum sweeping speed of not less than 5 mph at engine speeds necessary for efficient magnet operation.

3.6.2 Travel speed. Sweeper shall have a safe traveling speed of not less than 10 mph.

3.6.3 Turning radius. Minimum outside turning radius of the tractor shall be not more than 17 feet.

* 3.6.4 Pickup and residual magnetism. The sweeper magnet, set at an average clearance of 4 inches, shall be capable of picking up 98 percent of 50 pounds (lb) of miscellaneous small steel items such as washers, nuts, wire, flattened cans, spread completely under the entire magnet area. When release cycled, not less than 96 percent of the pickup load shall fall to the ground.

3.6.5 Altitude. The sweeper shall operate satisfactorily at actual or simulated altitudes ranging from sea level to 3,000 feet.

3.6.6 Slopes. The sweeper shall negotiate side slopes of 15 percent and longitudinal slopes of 20 percent without any wheel of the sweeper losing contact with the ground and without loss of fuel or lubricants.

3.6.7 Transport.

3.6.7.1 Road. The sweeper with the magnet assembly raised in travel position shall be capable of negotiating highway, secondary road, and cross-country road surfaces without damage or permanent deformation. The sweeper shall be capable of attaining a maximum speed of 10 mph and maintaining an average speed of 8.0 mph. The sweeper shall be capable of being towed at 15 mph using a military towbar conforming to MS500048 without interference or failure to track the towing vehicle. The sweeper shall be capable of self-load, transport at convoy speed, and unload from the standard 25 ton low-bed trailer.

* 3.6.7.2 Rail. The sweeper shall be capable of rail impact when tested in accordance with MIL-STD-810, method 516, procedure VI.

3.6.7.3 Air. The sweeper measurements shall conform to the dimensional and loading ramp limitations of a military C-130 aircraft (Figure 1) without disassembly, except that disconnecting and subsequent reconnection can be performed by one man in 1 hour.

3.6.8 Rain and wind protection. The sweeper shall be protected from damage due to all combinations of heavy rain and high wind during both limited and extended outside storage.

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3.6.9 Lifting and tying down attachments. Sweeper shall be equipped with lifting and tying down attachments. Attachments shall conform to MIL-STD-209, type II or type III. Attachments shall be indicated by a data plate, except tying down attachments may be identified by stenciling or other suitable marking on the sweeper. Tying down marking shall clearly indicate that the attachments are intended for tying down of the sweeper on the carrier when shipped. Data plates shall be of a material which will be durable, corrosion resistant, and remain legible for the life of the equipment, and shall be mounted in a conspicuous location. The backing of the plate and the surface to which the plate is applied shall be made corrosion resistant by application of a paint or preservative compound. Lifting attachment data plate shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the sweeper showing the center of gravity shall be provided on the data plate.

* 3.7 Tractor. Unless otherwise specified (see 6.2), the tractor shall be diesel-engine-driven and comply with KKK-T-640, size W-40, standard type with nonadjustable wheel treads. The diesel engine shall start in any temperature above -20° Fahrenheit (F). Starting aids may be either glow plug or ether primer. When ether priming system is furnished, it shall be of the measured shot type with storage capacity of not less than 12 fluid ounces. The tractor shall be provided with a drawbar-hitch, a rear power takeoff connection, and a tachometer. The basic tractor shall be as specified in 3.7.1 through 3.7.7.

3.7.1 Frame. The tractor frame shall be the manufacturer's commercial standard, reinforced and adapted as required to adequately accommodate and support the additional components without impairing the normal maneuverability of the tractor.

3.7.2 Wheels, rims, and tires. Wheels shall be the manufacturer's current commercial standard type, provided with ball or roller bearing hubs, and demountable at the hub or rim. Rim and tire ratings shall conform to the T&RA Yearbook recommendations for the type and size of tire furnished.

3.7.2.1. Tires. Tires shall have a rated capacity at least equal to the load imposed on each tire, measured at each wheel at the ground, with the sweeper fully equipped for operation. Tube or tubeless type tires shall be furnished.

3.7.2.2 Front wheel tires. The front wheel tires shall be of the puncture-proof type.

3.7.3 Steering. Manufacturer's standard power steering shall be furnished. Steering system shall permit manual steering in the event of failure of the power mechanism.

3.7.4 Speed regulation. A governor shall be provided for controlling engine speed within limits to insure that the generator output will not lower to such a degree that the magnet pickup load will fall to the ground. The governor shall be of a variable speed type and shall be operable throughout the range necessary to provide sweeping speed from 2.5 to 5 mph. Governor adjustment shall be automatic and shall regulate when throttle position is changed. A throttle shall be provided within convenient reach of the operator to permit selection of the desired tractor speed.



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* 3.7.5 Cab. When specified (see 6.2), the sweeper shall be provided with a steel cab of manufacturer's standard current type. The cab shall have safety plate glass windows, at least one door which shall be on the left or right-hand side of the cab. Windows shall be provided with adjustable openings for ventilation and have sufficient areas to allow unobstructed view to the front, rear, and both sides of the sweeper. Cab shall be furnished with the following accessories:

a. Windshield wiper, air or electrically powered.
b. Windshield washer.
c. Sun visors.
d. Heater and defroster.
e. Dome light or other means for cab lighting.
f. When specified (see 6.2), a rotating vehicular warning light or stationary strobe light with an aviation yellow dome shall be furnished. The warning light shall be suitable for mounting on top of the cab and be compatible with the tractor's electrical system.

3.7.6 Engine hour meter. An engine hour meter having a totalizing mechanism of not less than 9,999 hours shall be furnished to register accurately the number of hours of engine operating time. Meter shall be of strong rugged construction to insure continuous trouble-free performance, and shall be so mounted on the tractor that it can be conveniently and easily read by personnel servicing the tractor.

* 3.7.7 Brakes. Service brakes and parking brakes shall be provided. Braking system provided shall conform to the tractor performance requirements of SAE J1152.

3.8 Hydraulic system. The hydraulic system shall consist of rams, pump, supply tank, piping, fittings, valves, controls, and all other parts essential to efficient operation. The pump shall be driven by the tractor engine. All controls shall be so located as to be readily accessible to the operator from the driver's position.

3.8.1 Pump. The pump shall be of a heavy duty, positive or variable displacement type, incorporating ball or roller bearings.

* 3.8.2 Lines and fittings. Hydraulic tubes, hoses, and fittings shall conform to SAE J514, J516, J517, J518, and J524. A minimum number of fittings, joints, and connections shall be used to prevent excessive back pressure, clogging, and leakage.

3.8.3 Supply tank. Tank shall have sufficient capacity to hold the maximum volume of fluid, plus a minimum reserve of 20 percent required in the system. Tank shall be provided with an easily accessible filler opening. A vented filler cap, oil strainer, and shutoff valve shall be furnished. An oil strainer in the system may be used.

* 3.9 Maintainability. When specified (see 6.2), maintainability provisions shall be furnished. Ease of maintenance provisions shall incorporate features insuring operating tool and lubricant component clearances and shall facilitate maintenance and servicing operations at extremely low temperatures. Dimensions



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of hand access openings shall be in accordance with MIL-STD-1472. All fuel, lubricant, coolant, and liquid reservoirs shall drain to the ground. Each maintenance, assembly, or disassembly operation performed as a result of testing in accordance with 4.4.11 shall be accomplished by two men (hoist and hoist operator may be used as necessary) using common tools and special tools furnished with the sweeper. After completion of any maintenance operation, the sweeper shall operate as specified herein.

3.10 Magnet assembly.

* 3.10.1 Magnet. The magnet shall be of rugged construction capable of withstanding its intended hard usage under all conditions specified herein. The magnet shall be rated for 8 hours continuous operation at normal current load with a maximum temperature rise of 257°F in an ambient temperature of 104°F. The power requirements shall not exceed 4,000 watts after the rated voltage of 115 volts (V) have been applied for a period of 90 minutes in an ambient temperature of 77°F. Magnet coil shall use class H insulation and shall be capable of withstanding 1500 V alternating current applied between the coil terminals and main frame for a period of 1 minute without injury. Hot rolled steel plate or cast structure shall be used in the magnet frame. Construction shall be such as to prevent any movement of the components due to magnet action or physical shock. The magnet shall be a minimum of 8.0 foot long, 17 inches wide, and shall have an effective area of at least 1,700 square inches. A replaceable protective guard of nonmagnetic material shall cover the leading edge of the magnet and shall have sufficient strength to withstand any impact normally encountered in sweeping. The lower face of the magnet shall be covered by either laminated glass fabric complying to ASTM D709, grade G-5; or nonmagnetic corrosion-resistant steel having a minimum thickness of 0.0598 inch (US revised standard gage No. 16). The laminated glass fabric cover shall be at least 0.25 inches thick and shall be secured to the magnet by nonmagnetic bolts. The magnet shall pass the operational pickup test specified in 4.4.3.5 and the pickup and residual magnetism test specified in 4.4.3.6.

3.10.2 Magnet suspension. The magnet shall be suspended from the chassis frame between the front and rear axles with the 8.0 foot length mounted parallel to the axles. The leading edge of the magnet shall be 2 inches higher than the trailing edge. The design shall be such, that the installation and removal of the magnet assembly can be accomplished within 30 minutes. The removal of the magnet with no further modifications shall permit the operation of the unit as a tractor. The suspension shall be such as to permit the magnet to be raised and lowered hydraulically. Hydraulic rams with associated crank arms, or other mechanical means of good design shall be employed. The suspension shall be designed to prevent the magnet, when the sweeper is operating at 5 mph, from absorbing the full impact when hit by any obstruction up to 6 inches in height. The suspension shall cause the magnet to rise in an arc above the 6 inch obstruction following contact and to return to the original operating position after passing over the obstruction. Means shall be provided to prevent the magnet from swinging freely during normal operating conditions. The suspension shall be designed in a manner to withstand all normal shock loads multiplied by a factor of safety consistent with good design practices.



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3.10.3 Magnet height control. A control, operated from the driver's position, shall be provided to control the ground clearance of the magnet from 2 to 8 inches at the trailing edge. An indicator shall be provided to enable the ground clearance of the magnet to be readily determined from the operator's position.

3.11 Generator. The generator shall conform to NEMA MG-1 and MG-2, and shall be the manufacturer's standard for this application. It shall have a rating of not less than 5 kW output, 120 V dc.

3.11.1 Generator drive. The generator drive shall operate independently of the rear axle drive, permitting generator operation at all times during which the tractor's engine is running. Belt and pulley drive of a generator shall have a protective guard or be so positioned as to not constitute a safety hazard to the operator.

3.11.2 Generator clutch. A clutch shall be provided for disengaging the generator when desired. Direct mechanical drive with clutch release is acceptable. The generator shall not be disengaged during the shifting of any of the tractor's transmission gears during the sweeping operation.

3.12 Control box and wiring. The control box, control box components, and all electrical wiring internal and external to the control box shall conform to NFPA No. 70. Instrument meters shall be of a type which are sealed to prevent the entrance of dust and moisture. The control box or components shall be of such a type that dust and moisture will not impair the function of the control box components.

3.12.1 Control box. A weatherproof control box, suitable for its intended outdoor use, shall be provided and conveniently located near the driver's position. The control box shall contain the following components, suitable for the intended use:

a. Voltmeter 0-150 V dc.
b. Ammeter 0-50 amperes, dc.
c. Power indicator light.
d. Generator field rheostat or adjustable voltage regulator for voltage adjustment.
e. Main control circuit breaker with thermal overload trip.
f. Field discharge device.

3.12.2 Main control. The main control device shall be provided with a circuit breaker having thermal overload trip and necessary contact arrangement so that, when the circuit breaker trips, a field discharge device of adequate power handling capability and ruggedness shall instantly shunt the magnet winding terminals for protection against a high inductive voltage surge. The indicator light circuit shall be energized when the circuit breaker is closed.

3.12.3 Wiring and wiring diagram. Wiring and connections to components of the control box shall conform to good commercial practice. Wiring shall be arranged in a manner which shall facilitate removal of parts. A label showing the wiring diagram of all circuits and terminals shall be mounted inside the control box for easy reference, and shall be protected from abrasion and weather damage.



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3.13 Lubrication. Lubrication means shall be provided for all parts of the sweeper normally requiring lubrication. Where the use of high lubrication pressure will damage grease seals, fittings with pressure release shall be used.

3.14 Electromagnetic interference characteristics. When specified (see 6.2), the equipment shall conform to the electromagnetic characteristic requirements and test limits for class number IIIC equipment as specified in MIL-STD-461.

* 3.15 Cleaning, treatment, and painting. Unless otherwise specified, surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. When specified (see 6.2), the sweeper shall be cleaned, treated, and painted in accordance with MIL-T-704, type A. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects. When specified (see 6.2), the finish coat shall be in accordance with MIL-C-46168.

3.16 Identification plate. The contracting officer will furnish to the Government inspector the required identification plates. The contractor will be required to stamp the necessary data in the blank spaces thereon and securely affix said plates in a conspicuous place on each unit. Nonferrous screws, rivets, or bolts, of not less than 1/8-inch nominal diameter shall be used to affix the plates. Nomenclature shall be: "SWEEPER MAGNET, 40 HP, WHEEL, TRACTOR DRIVEN".

* 3.17 Instruction plates. The equipment shall be provided with instruction plates describing special or important procedures for operating and servicing equipment or warnings of hazardous procedures. The plates shall be durable and legible throughout the life of the equipment. The plates shall be conspicuously located. Nonferrous screws or bolts not less than 1/8 inch in diameter shall be used to affix the plates to the equipment.

3.18 Workmanship.

* 3.18.1 Steel fabrication. Steel used in the fabrication of equipment shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and chipping shall be done neatly and accurately. All bends of a major character shall be made with controlled means in order to insure uniformity of size and shape.



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3.18.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.18.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

* 3.18.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2.1).
(b) Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one sweeper when a first article is required (see 3.2). This inspection shall include the examination of 4.3 and the tests of 4.4.1 through 4.4.5; and when specified, the tests of 4.4.6, 4.4.7, 4.4.8, 4.4.10, and 4.4.11 (see applicable paragraphs in Section 4 and 6.2). The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.



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4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the test of 4.4.9, and the packaging inspection of 4.5.

4.3 Examination. Each sweeper shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements or any necessary redesign or modification following failure to meet specified requirements, shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements.

4.4 Tests.

4.4.1 Test conditions. Prior to testing, perform all routine service and maintenance operations. Inability to perform service and mainentance operations, drainage spilling over other components of the sweeper, difficulty in filling or inaccessibility for service shall constitute failure of this test. Only that maintenance established by the contractor and submitted as a maintenance schedule prior to start of testing shall be performed during test. Unless otherwise specified (see 6.2), the reliability test (4.4.10) shall be conducted on a 24-hour, 7-day-per-week continuous basis (with down-time for normal maintenance). Safety and human factor considerations shall be evaluated throughout testing.

4.4.2 Hydraulic system. The sweeper, after being serviced for run-in and the magnet installed and adjusted, shall be placed on a level paved surface. With clutch(s) engaged operating the generator and hydraulic pump, raise the magnet to the maximum raised position. Lower the magnet to the maximum lowest preset sweeping position. Repeat this operation through 30 complete cycles. Raise the magnet to the maximum raised position and measure and record the magnet height from paved surface to a fixed position on each front corner of the magnet. With tractor engine and generator operating, hold the magnet in the raised position for a period of 30 minutes and measure the height of the magnet from the same fixed position. Any evidence of hydraulic leaks, drift, or downward movement of the magnet in excess of 0.5 inch shall constitute failure of this test.

4.4.3 Performance test.

4.4.3.1 Service tests. Sweeper shall be run-in as prescribed by the tractor manufacturer. After run-in, the following tests shall be performed with the fully equipped sweeper.

4.4.3.2 Speed tests. Speed tests in all gears shall be conducted over a section of not less than 250 feet of straight, level concrete. Two or more runs shall be made in each gear. The average speed in each gear shall be recorded. Nonconformance to the requirements of 3.6.1, 3.6.2, or the tractor manufacturer's given values shall constitute failure of this test.

4.4.3.3 Brake tests. Sweeper, with magnet installed, shall be driven down a paved road with a minimum of 5 percent grade at 9 mph. Service brakes shall be activated and the distance between points of activation and stopping shall be measured. The distance shall be not more than 25 feet. Parking brake shall then be set and locked. Evidence of brake slippage shall constitute failure of this test.

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4.4.3.4 Road test. All adjustments for normal magnetic sweeping shall be made. The sweeper shall be driven over a roadway for a distance of 15 miles at a speed of 5 mph. Generators shall supply normal output to the magnet. The sweeper shall be stopped once in every mile. During the run, the sweeper shall be subjected to sharp U-turns and figure eights, in all gears. Any impairment to the normal maneuverability of the tractor shall constitute failure of this test.

4.4.3.5 Operational pickup test. Within 10 minutes after completion of the road test (see 4.4.3.4), the sweeper shall be set for an average clearance of 4 inches above a slab of open hearth, mild steel plate. The slab shall weigh 500 lb ±5 percent and shall be a minimum of 7.5 feet long, 27 ±1 inch wide, and 1 ±0.25 inch thick. The magnet shall pick up the slab from the ground with the sweeper moving forward at a speed between 2.5 and 5 mph in a direction perpendicular to the length of the slab. The slab shall be retained by the magnet as the speed of the sweeper is varied from 2.5 to 5 mph and during the process of shifting all transmission gears of the tractor while in motion. Upon completion of the test, the equipment shall be inspected for evidence of any mechanical or electrical defects which would constitute failure of this test.

4.4.3.6 Pickup and residual magnetism test. Sweeper shall be tested to verify conformance to 3.6.4. Any evidence of nonconformance to 3.6.4 shall constitute failure of this test.

4.4.3.7 Dielectric test. The magnet of the sweeper shall be subjected to a voltage of 1500 V root mean square, 60 Hertz between the winding and the core of the magnet for a period of 1 minute. Any failure or damage to the insulation shall constitute failure of this test.

4.4.4 Power test. The magnet shall be tested to determine conformance to the power requirements of 3.10.1.

4.4.5 Slopes. Drive the sweeper up and down a 20 percent longitudinal slope. Drive the sweeper on a 15 percent side slope. Check stability. Failure to operate properly shall constitute failure of the test.

* 4.4.6 Rain. When specified (see 6.2), a rain test shall be conducted. Subject the sweeper to a simulated rainfall of 4.0 ±1 inch per hour as measured by a US Weather Bureau type gage. Simulated rainfall shall be in the form of droplets having a maximum diameter of 0.060 inches. The simulated rainfall shall fall on the sweeper impinging with a force equivalent to a 20 mph wind. The simulated rainfall shall disperse uniformly over the surface of the sweeper simultaneously for not less than 30 minutes on each surface. The engines shall be started after the rain test and idle for 15 minutes. Evidence of water penetration such as free water in the machinery area, inability of the engines to start and idle, or inability to perform sweeper functions shall constitute failure of this test.

* 4.4.7 Transportability. When specified (see 6.2), a transportability test shall be conducted. Place the sweeper in position for shipment. Measure sweeper. Compare dimensions with clearance diagrams for C130 aircraft. Perform a rail impact test in accordance with MIL-STD-810, method 516, procedure VI. Test lifting and tiedown attachments in accordance with MIL-STD-209.



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* 4.4.8 Sound level. When specified (see 6.2), a sound level test shall be conducted. The sound level at the operator's station shall be measured in accordance with SAE J919. The exterior noise level shall be measured in accordance with SAE J88. Nonconformance to 3.5.3.2 shall constitute failure of this test.

4.4.9 Operational test.

4.4.9.1 Adjustment of controls. The controls shall be moved through all possible positions to determine ease of adjustment. There shall be no binding or scraping in any position.

4.4.9.2 Pickup and drive test. Sweeper shall be subjected to the operational pickup test in accordance with 4.4.3.5 after the tractor's engine has been running and the magnet energized for 90 minutes. The 500 lb load shall then be released and the sweeper shall be subjected to sharp U-turns and figure-eight turns in all transmission gears to determine that the normal maneuverability of the tractor is not impaired. The sweeper shall suffer no mechanical and electrical defects.

4.4.10 Reliability. When specified (see 6.2), a reliability test shall be performed. The initial production sweeper shall be run-in for not less than 10 hours prior to start of reliability testing. The tests shall consist of conducting sweeping operations on airfields, and paved and secondary roads. After each 25 hours of sweeping operations, the "operational pickup test" specified in 4.4.3.5 and the "pickup and residual magnetism test" specified in 4.4.3.6 shall be conducted. Test operations shall continue for a sufficient length of time to reach a decision to "accept" or "reject" in accordance with MIL-STD-781, test plan II. Failure, for the purpose of this test, is defined as any malfunction which the operator cannot remedy by adjustment, repair, or replacement action using controls, on-equipment tools, and on-equipment parts within 1 hour and which causes or may cause:

a. Failure to commence operation, cessation of operation, or degradation of performance capability of the sweeper below designated levels.
b. Serious damage to the sweeper by continued operation.
c. Serious personnel safety hazards.

4.4.11 Maintainability. When specified (see 6.2), a maintainability test shall be performed. Maintenance operations required during initial production testing shall be accomplished in accordance with the maintenance ratio specified in 3.9. Nonconformance to 3.9 shall constitute failure of this test. During initial production testing, maintenance support elements such as manuals, lubrication charts, repair parts, and special tools (when required by the contract) shall be evaluated. Inability of the sweeper to be properly maintained, utilizing the support elements, shall constitute failure of this test and shall require revision of the document(s). Errors or inadequacies in the manuals shall not be considered in assessing the maintenance ratio or the mean-time-between-failure.

4.5 Packaging inspection. The preservation, packing, and marking of the sweepers shall be inspected to verify conformance to the requirements of section 5.



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5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Commercial.

5.1.1.2 Cleaning and drying. Prior to the application of preservative compounds or paint, surfaces shall be cleaned by process C-1 and dried by any applicable procedure of MIL-P-116.

5.1.1.3 Tractor and hydraulic system. The tractor, hydraulic system, repair parts, tools, and publications shall be preserved and package in accordance with the level A, type II, unboxed (mobile) requirements of MIL-T-3351.

5.1.1.4 Magnet. The magnet shall be removed from the tractor as preserved method I. All unpainted exposed metal surfaces of the magnet, including threaded surfaces and surfaces exposed by disassembly, shall be coated with type P-1 preservative.

5.1.1.5 Generator. The magnet generator shall be preserved and packaged in accordance with level A of MIL-E-16298, using the alternate method specified for electric machines attached to mechanical equipment.

5.1.2 Commercial. The equipment shall be preserved in accordance with the contractor's standard practice in a manner to prevent deterioration and damage.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. The complete sweeper, except the magnet, shall be packed in accordance with level A, type II, unboxed (mobile) requirements of MIL-T-3351. The magnet shall be packed in a crate conforming to MIL-C-104, type II, class 1 or 2, style a, the contents shall be anchored, blocked, and braced in accordance with the crate specification.

5.2.2 Level B. The complete sweeper shall be packed as specified for level A, except the magnet shall be packed in a crate conforming to MIL-C-52950, type V, style A, class 1.

5.2.3 Commercial. The equipment shall be prepared for shipment in a manner which will insure arrival at destination in a satisfactory condition. Preparation for delivery shall comply with applicable carrier rules and regulations.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.



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6. NOTES

6.1 Intended use. Sweepers covered by this specification are intended for picking up ferrous object from runways, roads, and maintenance areas. The tractor may also be used for towing rotary brooms and other equipment supplemental to magnetic sweeping.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When first article is required for inspection and approval, and the time frame required (see 3.2, 4.2.1, and 6.4).
c. When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examinations and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
d. When the tractor required is other than specified (see 3.7).
e. When cab is required (see 3.7.5).
f. When warning light or strobe light is required (see 3.7.5 (f)).
g. When maintainability provisions are required (see 3.9).
h. When electromagnetic interference suppression is required (see 3.14).
i. When cleaning, treatment, and painting is other than specified (see 3.15).
j. When the sweeper shall be cleaned, treated, and painted in accordance with MIL-T-704, type A (see 3.15).
k. Color of finish coat of paint (see 3.15).
l. When final paint coat shall conform to MIL-C-46168 (see 3.15).
m. How reliability test shall be conducted if other than as specified (see 4.4.1).
n. When the rain test shall be performed (see 4.4.6).
o. When the transportability test shall be performed (see 4.4.7).
p. When sound level test shall be performed (see 4.4.8).
q. When reliability test shall be performed (see 4.4.10).
r. When maintainability test shall be performed (see 4.4.11).
s. Level of preservation and level of packing required (see 5.1 and 5.2).

* 6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

* 6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one complete sweeper. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.



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6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

6.6 Sound level. When the sound level of the equipment exceeds 85 dB(A) (re 20 MPa) at the operator's station, the appropriate discussion of the noise hazard per MIL-STD-1474 shall be conducted. The discussion shall include the requirement for hearing protection, type of hearing protection, the distance from which 85 dB(A) (re 20 MPa) will always be met, and what operational situatios or system configuration will reduce the level to 85 dB(A) (re 20 MPa).

Custodians:

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Navy - YD

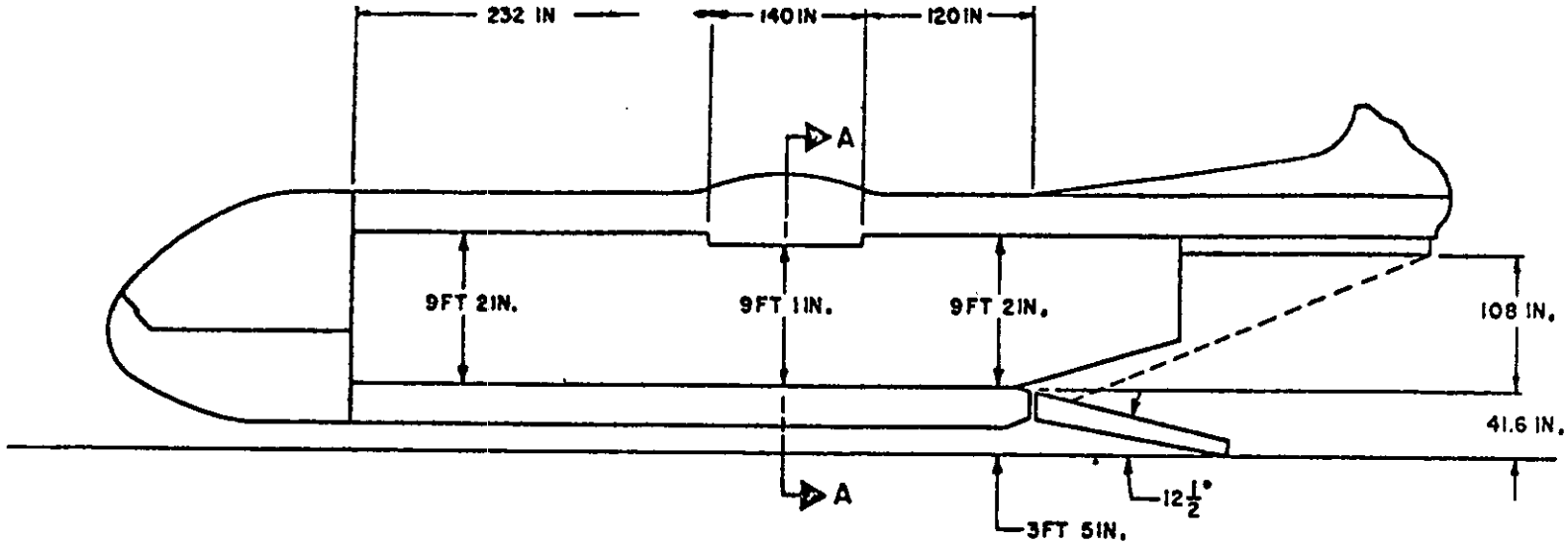
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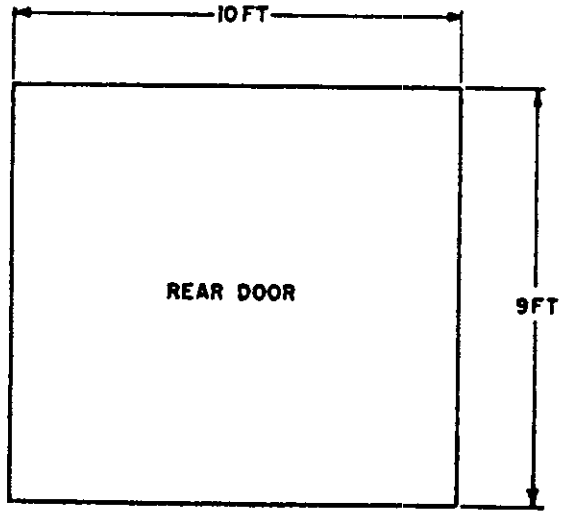
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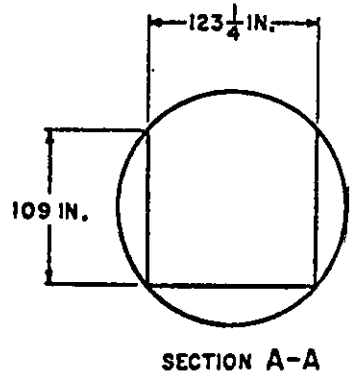




FIGURE 1. Cargo compartment profile (C-130).



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